PV101 Day 7 Student Print Packet

Compiled packet

Day_07_PV101_Load_Assessment_and_System_Sizing_2026-07-15

Contents follow in instructional order.

PV101 Day 7 Morning Quiz

Student copy | 10-15 minutes

Day_07_PV101_Load_Assessment_and_System_Sizing_2026-07-15

Name: _____ Date: _____

Answer briefly. Show work and units where needed.

1. What does AHJ stand for in a permitting context?

_____

_____

2. Why should the AHJ and utility requirements be checked before an installation begins?

_____

_____

3. Why must an incentive be verified with a current authoritative source?

_____

_____

4. Name two useful authoritative resources for planning permitted PV work.

_____

_____

5. What is the purpose of a permit and inspection process?

_____

_____

6. What should you do first when a practice question includes units?

_____

_____

7. When two answers seem possible, what type of answer takes priority in PV work?

_____

_____

8. Why should a calculated answer be checked for reasonableness?

_____

_____

9. Why is reviewing a missed practice question useful?

_____

_____

10. State the course rule that replaces guessing.

_____

_____

PV101 Day 7 Student Worksheet

Load assessment and basic system sizing | Guided practice

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Name: _____ Date: _____

Operating rules

- Use only the values provided.
- Show formulas, numbers, and units.
- Circle assumptions.
- Check reasonableness before asking for signoff.
- This is guided practice—not a product design or proposal.

Provided training assumptions

Item	Training value
System voltage	12 V
Usable battery fraction / maximum depth of discharge	80% for this exercise
Autonomy for this exercise	1 day
Peak sun hours for this exercise	4 h/day
Loss multiplier for this exercise	1.25
Inverter headroom for this exercise	25% above concurrent AC load

Part A: RV training loads

Load	Power	Hours/day	Calculation	Wh/day
3 interior LED lights	8 W each	4 h		
Water pump	60 W	0.5 h		
Vent fan	36 W	3 h		

Phone/USB charging	24 W	2 h
Laptop charger (AC)	65 W	2 h

Total daily energy: _____ Wh/day

Part B: Storage

Required stored Wh = daily Wh x autonomy = _____

_____

Battery nameplate Wh = required stored Wh / usable fraction = _____

_____

Battery Ah at 12 V = battery nameplate Wh / 12 V = _____

_____

Reasonableness sentence: _____

_____

Part C: Array and inverter

Raw array watts = daily Wh / peak sun hours = _____ W

_____

Adjusted array watts = raw array watts x 1.25 = _____ W

_____

Concurrent AC load used for inverter sizing: _____ W

_____

Minimum inverter rating with 25% headroom: _____ W

_____

What load or assumption most affects the result? _____

_____

Part D: One-line diagram

Draw: PV -> controller -> battery -> inverter/DC loads. Show return paths, protection, disconnects, labels, and meter points.

_____

_____

_____

_____

Verification and signoff

Check	Peer	Instructor
Loads and hours copied correctly		
Every calculation shows units		
Assumptions match provided values		
Answers are reasonable and rounded clearly		
Diagram includes both directions/return path		
Protection, disconnects, labels, and test points shown		

PV101 Day 7 Guided-Practice Guide

Student-facing

Day_07_PV101_Load_Assessment_and_System_Sizing_2026-07-15

Name: _____ Date: _____

Purpose

Guide students through a bounded sizing exercise using only the provided RV training loads and assumptions.

Tools and materials

- Student worksheet
- Calculator and pencil
- Ruler
- Provided training assumptions
- No physical PV equipment required

Procedure

- Complete the loads table.
- Checkpoint 1: instructor verifies daily Wh.
- Calculate storage using the provided autonomy and 80% usable-fraction / maximum-depth-of-discharge assumption.
- Checkpoint 2: instructor verifies units and arithmetic.
- Calculate raw and adjusted array watts.
- Select inverter headroom from concurrent AC load.
- Draw and peer-check the one-line diagram.
- Revise and obtain instructor signoff.

Common mistakes

- Adding watts without multiplying by hours
- Confusing W with Wh
- Dividing by the wrong voltage
- Omitting the usable-fraction / depth-of-discharge adjustment
- Sizing inverter from daily energy instead of simultaneous power
- Inventing equipment values not supplied

Cleanup

- Return calculators and rulers.
- Submit worksheet and reflection.
- Clear tables and recycle scratch paper as directed.

PV101 Day 7 Reflection Questions

Student copy

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Name: _____ Date: _____

1. What sizing step makes the most sense to you now?

_____

_____

2. Where are you most likely to make a unit or assumption error?

_____

_____

3. How does verification change a design calculation?

_____

_____

4. What did Days 1-6 teach that helped today?

_____

_____

5. What would you check before trusting someone else's sizing worksheet?

_____

_____